

Day - 05

Registers & Counters - Assignment

1. Design a counter with T flipflops whose count sequence is
0, 2, 4, 5, 3, 2, 0, ...

Soln

Let;

Current State			Next State			Result		
Q_2	Q_1	Q_0	Q_2	Q_1	Q_0	T_2	T_1	T_0
0	0	0	0	1	0	0	1	0
0	0	1	0	0	0	0	0	1
0	1	0	1	0	0	1	1	0
0	1	1	0	0	1	0	1	0
1	0	0	1	0	1	0	0	1
1	0	1	0	1	1	1	1	0
1	1	0	X	X	X	X	X	X
1	1	1	X	X	X	X	X	X

T_0 ;

$Q_2 \backslash Q_1 Q_0$	00	01	11	10
0	0	1	0	0
1	1	0	X	X

$$T_2 = Q_2 \bar{Q}_0 + \bar{Q}_2 \bar{Q}_1 Q_0$$

T_1 ;

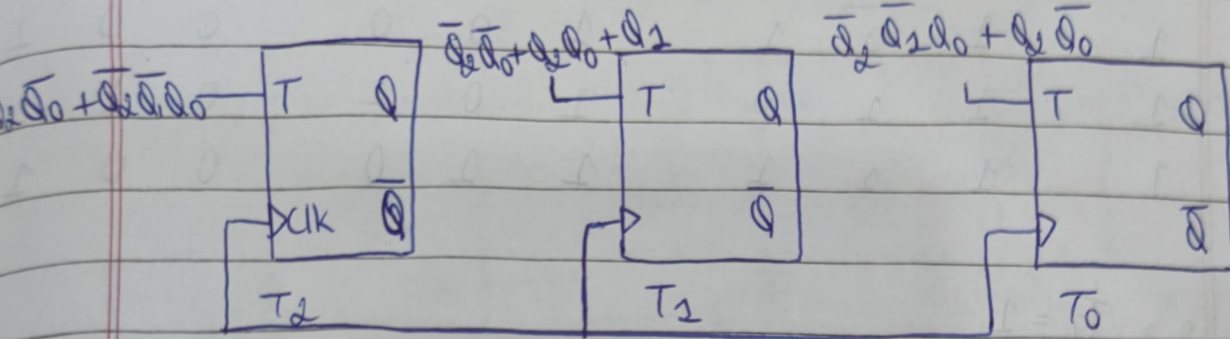
$Q_2 \backslash Q_1 Q_0$	00	01	11	10
0	1	0	1	1
1	0	1	X	X

$$T_1 = \bar{Q}_2 \bar{Q}_0 + Q_2 Q_0 + Q_2 Q_1 Q_0$$

T_0 ; $Q_2 \backslash Q_1 Q_0$

	00	01	10	11
0	0	1	0	0
1	1	0	X	X

$$T_0 = \bar{Q}_2 \bar{Q}_1 Q_0 + Q_2 \bar{Q}_0$$



2. Design a 3 bit updown counter using negative edge triggered T-FF.

Soln

Input <u>M</u>	Present State			Next State			T Flip Flop Input		
	<u>Q_2</u>	<u>Q_1</u>	<u>Q_0</u>	<u>Q_2^+</u>	<u>Q_1^+</u>	<u>Q_0^+</u>	<u>T_2</u>	<u>T_1</u>	<u>T_0</u>
0	0	0	0	0	0	1	0	0	1
0	0	0	1	0	1	0	0	1	1
0	0	1	0	0	1	1	0	0	1
0	0	1	1	1	0	0	1	1	1
0	1	0	0	1	0	1	0	0	1
0	1	0	1	1	1	0	0	1	1
0	1	1	0	1	1	1	0	0	1
0	1	1	1	0	0	0	1	1	1
1	0	0	0	1	1	1	1	1	1
1	0	0	1	0	0	0	0	0	1
1	0	1	0	0	0	1	0	1	1
1	0	1	1	0	1	0	0	0	1

Input	Present State			Next State			T Flip Flop input		
M	Q ₂	Q ₁	Q ₀	Q ₂ ⁺	Q ₁ ⁺	Q ₀ ⁺	T ₂	T ₁	T ₀
1	1	0	0	0	1	1	1	1	1
1	1	0	1	1	0	0	0	0	1
1	1	1	0	1	0	1	0	1	1
1	1	1	1	1	1	0	0	0	1

∴ T₀ = 1

$$T_1 = M Q_2$$

	$Q_1 Q_0$	00	01	11	10
00	0	1	1	0	0
01	0	1	1	0	0
11	1	0	0	1	1
10	1	0	0	1	1

$$\therefore T_1 = \overline{M} \overline{Q}_0$$

$$\therefore T_1 = \overline{M} \overline{Q}_0 + M \overline{Q}_0$$

$$T_2 = M Q_2$$

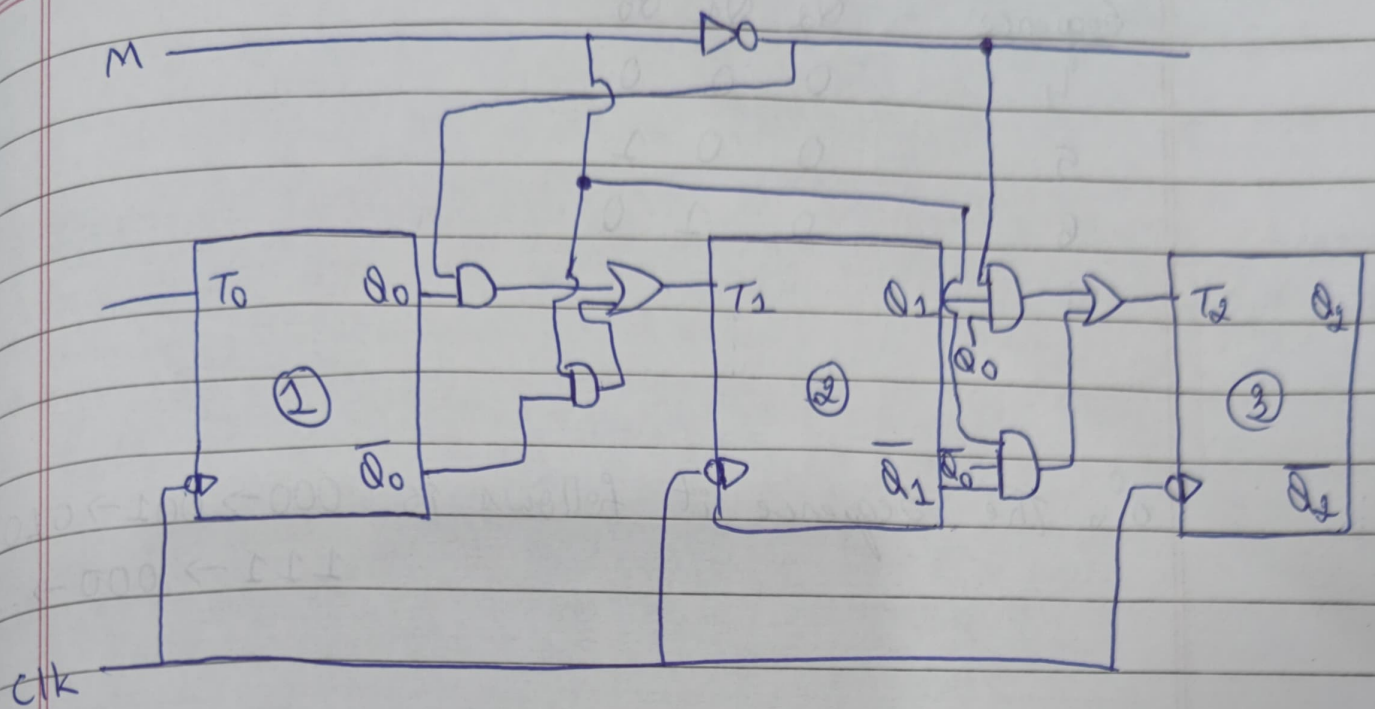
	$Q_1 Q_0$	00	01	11	10
00	0	0	1	0	
01	0	0	1	0	
11	1	0	0	0	
10	1	0	0	0	

$$\therefore T_2 = \overline{M} Q_1 Q_0 + M \overline{Q}_1 \overline{Q}_0$$

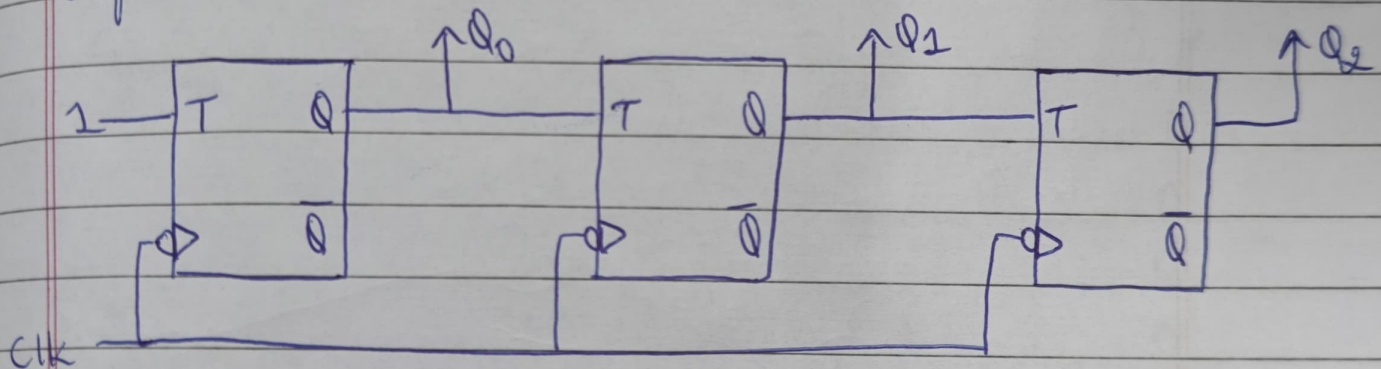
∴ T₀ = 1

$$T_1 = \overline{M} \overline{Q}_0 + M \overline{Q}_0$$

$$T_2 = \overline{M} Q_1 Q_0 + M \overline{Q}_1 \overline{Q}_0$$



3. The circuit in the figure below is a counter. What is the sequence that this circuit counts in?



Soln These are negative edge triggered T flip flop
Assuming initial state $Q_2 Q_1 Q_0 = 000$

Sequence	Q_2	Q_1	Q_0
0	0	0	0
1	0	0	1
2	0	1	0
3	1	1	1

Sequence	Q_2	Q_1	Q_0
4	0	0	0
5	0	0	1
6	0	1	0
⋮	⋮	⋮	⋮

0°. The sequence it follows is $000 \rightarrow 001 \rightarrow 010 \rightarrow 111 \rightarrow 000 \rightarrow \dots$